



Student Number

2026 Year 12 Examination

Mathematics Extension 1

Trial examination

Date

17/08/2026

General

Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using blue or black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your NESA student number at the top of this page
- No white-out may be used

Total Marks:

70

Section I - 10 marks (pages 3 – 7)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II - 60 marks (pages 8 – 15)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

This question paper must not be removed from the examination room.

This assessment task constitutes 30% of the course.

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section.

Use the multiple-choice sheet for Question 1 – 10.

1. Consider the function $f(x) = e^x - 1$ and its inverse function $f^{-1}(x)$.

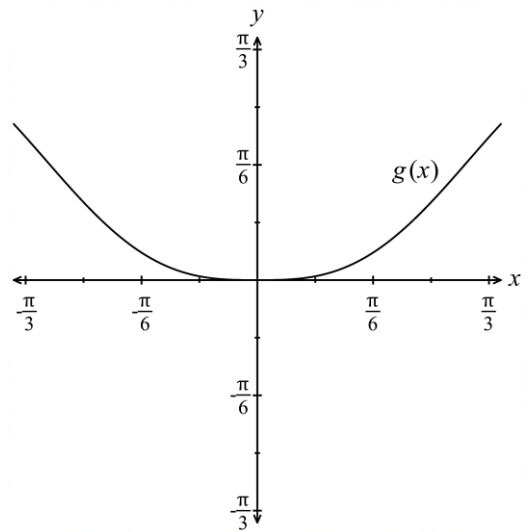
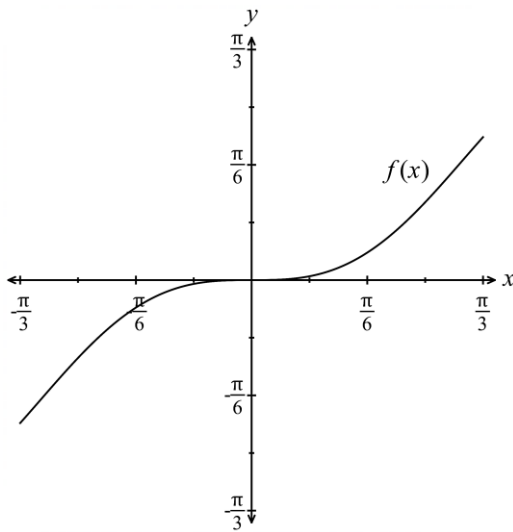
State the domain of the inverse function.

- A. $x \in (-\infty, \infty)$
 - B. $x \in (0, \infty)$
 - C. $x \in (-1, \infty)$
 - D. $x \in (-\infty, -1) \cup (-1, \infty)$
2. The polynomial $P(x) = x^3 - 4x^2 - 6x + k$ gives a remainder of 16 when divided by $x - 2$.

What is the value of k ?

- A. 4
- B. 12
- C. 20
- D. 36

3. The diagrams below show $f(x)$ and $g(x)$ respectively.



The function $g(x)$ may be written in terms of $f(x)$.

Which is the correct answer from the following options?

- A. $g(x) = f(|x|)$
 - B. $g(x) = f(-x)$
 - C. $g(x) = f^{-1}(x)$
 - D. $g(x) = f^{-1}(|x|)$
4. The coefficient of the x^2 term in the expansion of $(2x + 1)^n$ is 60.

What is the value of n ?

- A. 2
- B. 4
- C. 6
- D. 8

5. Consider the following integral,

$$\int \frac{4}{\sqrt{9 - 4x^2}} dx$$

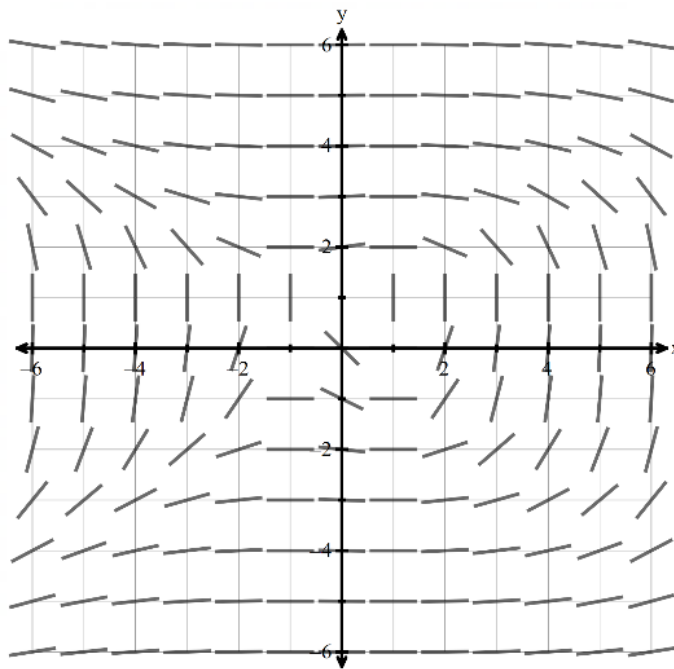
Which of the following expression is equal to the integral above?

- A. $2 \sin^{-1} \left(\frac{2x}{3} \right) + C$
 - B. $4 \sin^{-1} \left(\frac{2x}{3} \right) + C$
 - C. $2 \sin^{-1} \left(\frac{4x}{9} \right) + C$
 - D. $4 \sin^{-1} \left(\frac{4x}{9} \right) + C$
6. It is known that 36% of the customers of a certain supermarket will bring their own shopping bags. There are 3 cashiers and each cashier has 5 customers in queue.

What is the probability that among all the customers in queue, at least 3 of them have brought their own shopping bag.

- A. 0.0528
- B. 0.2501
- C. 0.7491
- D. 0.9472

7. Which of the following is a possible differential equation for the given slope field?



A. $\frac{dy}{dx} = \frac{y^3 - 1}{1 - x^2}$

B. $\frac{dy}{dx} = \frac{y^3 - 1}{x^2 - 1}$

C. $\frac{dy}{dx} = \frac{x^2 - 1}{y^3 - 1}$

D. $\frac{dy}{dx} = \frac{1 - x^2}{y^3 - 1}$

8. Alice, Beatrice, Charles, David and six other people are to be seated at a round table.

Alice and Beatrice must be seated directly opposite each other. Charles and David must also be seated directly opposite each other.

How many distinct seating arrangements are possible?

- A. $4 \times 6!$
- B. $4 \times 7!$
- C. $8 \times 6!$
- D. $8 \times 7!$
9. Which of the following integrals is equal to zero?

A. $\int_{-5}^5 (|x| - 2) \sin^{-1}(2x) \, dx$

B. $\int_{-5}^5 \sqrt[3]{x} \sin^{-1}(3x) \, dx$

C. $\int_{-5}^5 (|x| - 2) \cos^{-1}(2x) \, dx$

D. $\int_{-5}^5 \sqrt[3]{x} \cos^{-1}(3x) \, dx$

10. Let $\tilde{a}$ and $\tilde{b}$ be two different unit vectors such that $\tilde{a} + \tilde{b}$ is also a unit vector.

What is the value of $|\tilde{a} - \tilde{b}|$?

- A. 0
- B. $\frac{1}{\sqrt{2}}$
- C. $\sqrt{3}$
- D. $\sqrt{5}$

End of Section I

Section II

60 marks

Attempt Questions 11 - 14

Allow about 1 hour and 45 minutes for this section.

Answer each question in the given appropriate writing booklet. Extra writing booklets are available.

In questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

- (a) Consider the following function for $x \leq 4$. 2

$$f(x) = (x - 4)^2 - 3$$

Find the inverse function, $f^{-1}(x)$.

- (b) In how many ways can the letters of the word KILLARA be arranged so that the vowels are in alphabetical order but not necessarily together. 2

- (c) Differentiate $y = (\cos^{-1}(4x^2))^2$ 2

- (d) The acute angle between the vectors $\tilde{u}$ and $\tilde{v}$ is 60° . 2

The vector projection of $\tilde{u}$ onto $\tilde{v}$ is $4\tilde{i} - 3\tilde{j}$ and $|\tilde{v}| = 10$. Find $\tilde{v}$.

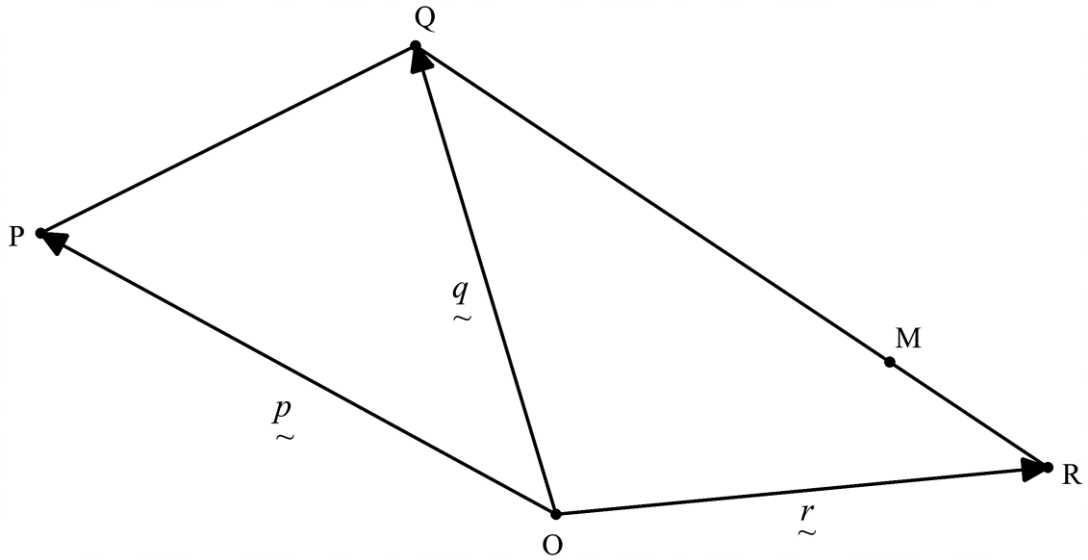
- (e) Find: 2

$$\int \sin^2(3u + 1) du$$

Question 11 continues on page 9

Question 11 (continued)

- (f) In quadrilateral $OPQR$ shown below, M lies on QR so that $|\overrightarrow{QM}| = 3|\overrightarrow{MR}|$



If $\overrightarrow{OP} = \underset{\sim}{p}$, $\overrightarrow{OQ} = \underset{\sim}{q}$ and $\overrightarrow{OR} = \underset{\sim}{r}$. Express the following in terms of $\underset{\sim}{p}$, $\underset{\sim}{q}$ and $\underset{\sim}{r}$

- | | | |
|------|-----------------------|---|
| i. | $\overrightarrow{PR}$ | 1 |
| ii. | $\overrightarrow{RM}$ | 2 |
| iii. | $\overrightarrow{PM}$ | 2 |

End of Question 11

Question 12 (14 marks)

- (a) A pie is placed in a preheated oven maintained at a constant temperature of 200°C . When the pie was placed into the oven, it had an internal temperature of 20°C .

Consider the following differential equation,

$$\frac{dP}{dt} = k(200 - P)$$

where P is the temperature of the pie, t minutes after it was placed into the oven and k is a positive constant.

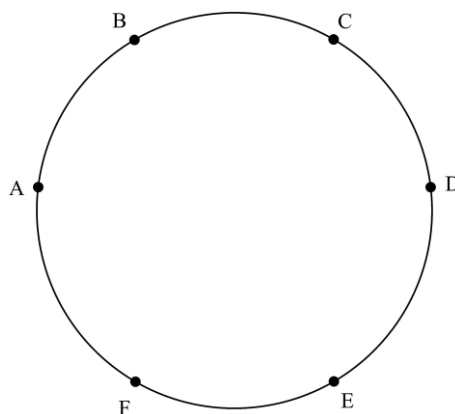
- i. By solving the differential equation, show that $P = 200 - 180e^{-kt}$. 2
- ii. The internal temperature of the pie reached 53°C after 10 minutes. Find the internal temperature of the pie after 20 minutes to the nearest degree. 2

- (b) Consider the following function $f(x)$.

$$f(x) = 3 \sin\left(x + \frac{\pi}{2}\right) - 4 \cos\left(\frac{\pi}{3} - x\right)$$

- i. Express the function in the form of $R \cos(x + \alpha)$. Give α correct to one decimal place where $0 < \alpha < \frac{\pi}{2}$. 3
- ii. Hence find the value of x , correct to 2 decimal places, at which $f(x)$ is at its minimum for $0 \leq x \leq 2\pi$. 2

- (c) Six points on a circle are given below. 2



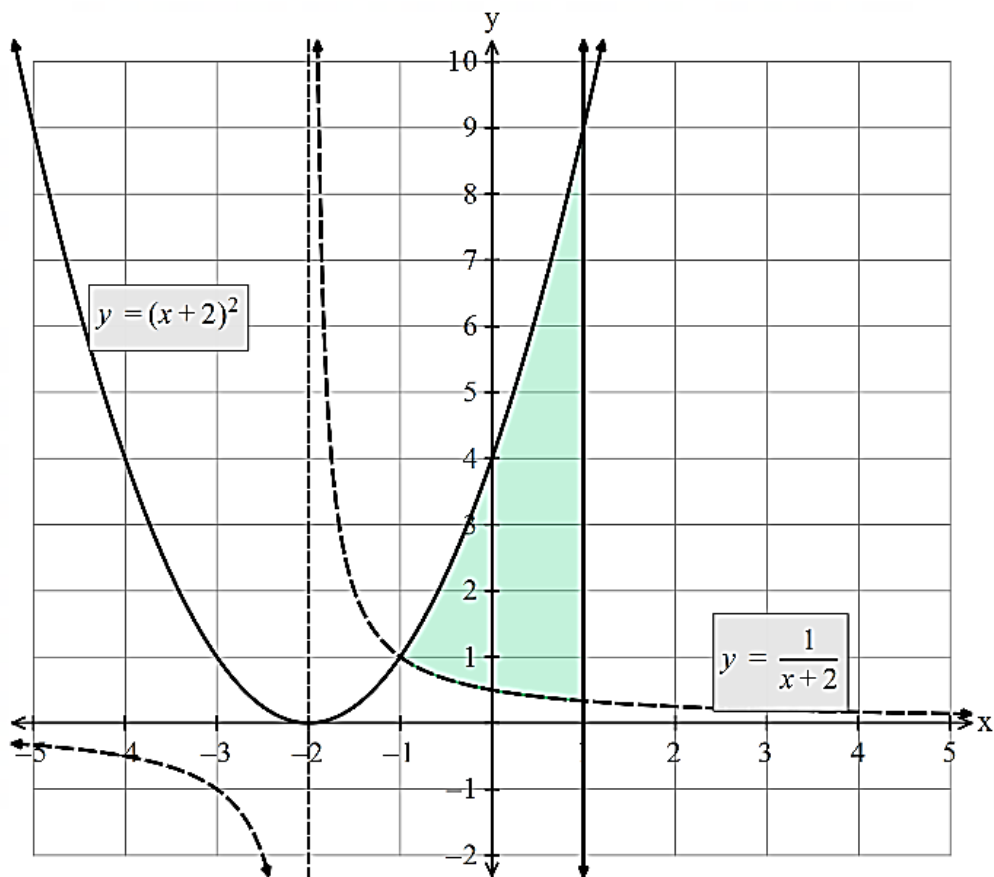
Four of the chords joining pairs of the six points are selected at random. What is the probability that the four chords form a quadrilateral?

Question 12 continues on page 11

Question 12 (continued)

- (d) Consider the shaded region below. It is bounded by $y = \frac{1}{x+2}$, $y = (x+2)^2$ and $x = 1$.

3



Find the exact volume of the solid formed by rotating the shaded area about x - axis.

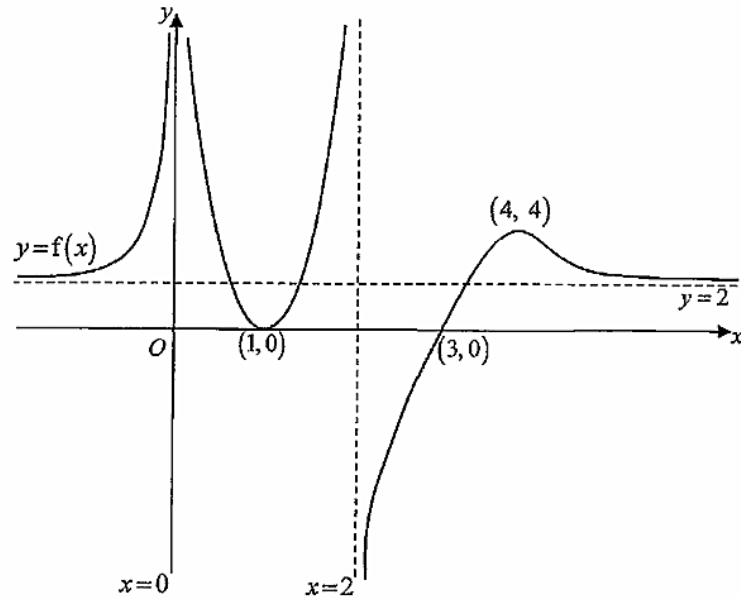
End of Question 12

Question 13 (14 marks)

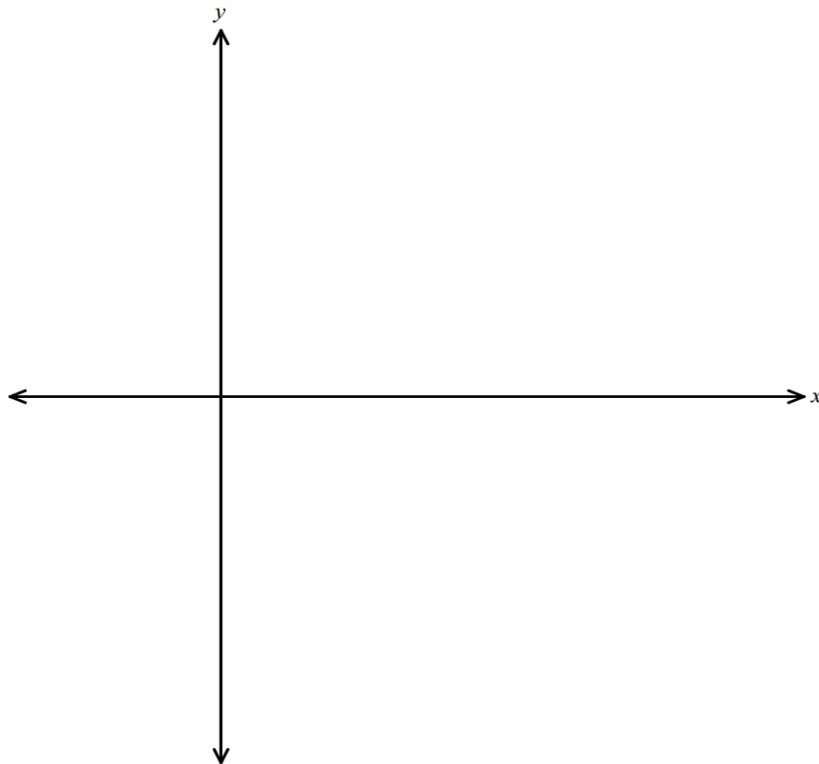
- (a) The diagram below shows a sketch of the graph of $y = f(x)$.

3

The curve passes through the points with coordinates $(1, 0)$ and $(3, 0)$, and has turning points at $(1, 0)$ and $(4, 4)$. The asymptotes are $x = 0$, $x = 2$ and $y = 2$.



Using the graph template provided below, sketch the graph of $y = \frac{1}{f(x)}$.



Question 13 continues on page 13

Question 13 (continued)

- (b) Show the following: 2

$$\left(1 + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \cdots + \binom{n}{n}\right)^2 = 1 + \binom{2n}{1} + \binom{2n}{2} + \cdots + \binom{2n}{2n}$$

- (c) Use the substitution $u = \sqrt{x}$ to show that 3

$$\int_0^{16} \frac{x}{1 + \sqrt{x}} dx = A - B \ln 5$$

Where A and B are positive constant values.

- (d) Data from the World Health Organisation showed that 25% of newborn babies have a birth weight of 3.8 kg or less.

A sample of 1000 Australian newborn babies were randomly selected.

Let $\hat{p}$ be the sample proportion of Australian newborn babies who had a birth weight of 3.8 kg or less.

- i. Assuming the sample proportion is normally distributed, show that the standard deviation of newborn babies in Australia is 0.0137. 1
- ii. Use the standard normal distribution table on **page 16**, approximate the probability that of the 1000 Australian newborn babies, at least 230 of them have a birth weight of 3.8 kg or less. 2
- (e) Given a polynomial $P(x) = x^3 + ax^2 + bx + c$, where a, b, c are non-zero constants. 3

The three roots of the polynomial forms the first three terms of a geometric sequence.

Show that,

$$b^3 = ca^3$$

End of Question 13

Question 14 (17 marks)

- (a) There are 8 different positive integers, none greater than 15. 2

By referring to the Pigeonhole Principle, show that at least three pairs of them have the same positive difference.

- (b) A bag contains counters, some of which are green and it is known that the probability of selecting a green counter is greater than 0.5. 4

One hundred trials of an experiment are run. In each trial, one counter is selected from the bag at random and its colour is noted. The counter is returned to the bag after each trial.

Let X be the random variable representing the number of times that a green counter is selected.

Given that $\text{Var}(X) = 16$ and $P(X \geq k) = 0.0668$. Find the value of k . You may use the standard normal approximation and the information on **page 16**.

- (c) (i) Let $t = \tan\left(\frac{x}{2}\right)$, prove the following identity. 2

$$\frac{1}{2} \cot\left(\frac{x}{2}\right) - \cot x = \frac{1}{2} \tan\left(\frac{x}{2}\right)$$

- (ii) Prove the following by mathematical induction for all $n \geq 1$ 3

$$\frac{1}{2} \tan\left(\frac{x}{2}\right) + \frac{1}{4} \tan\left(\frac{x}{4}\right) + \frac{1}{8} \tan\left(\frac{x}{8}\right) + \cdots + \frac{1}{2^n} \tan\left(\frac{x}{2^n}\right) = \frac{1}{2^n} \cot\left(\frac{x}{2^n}\right) - \cot x$$

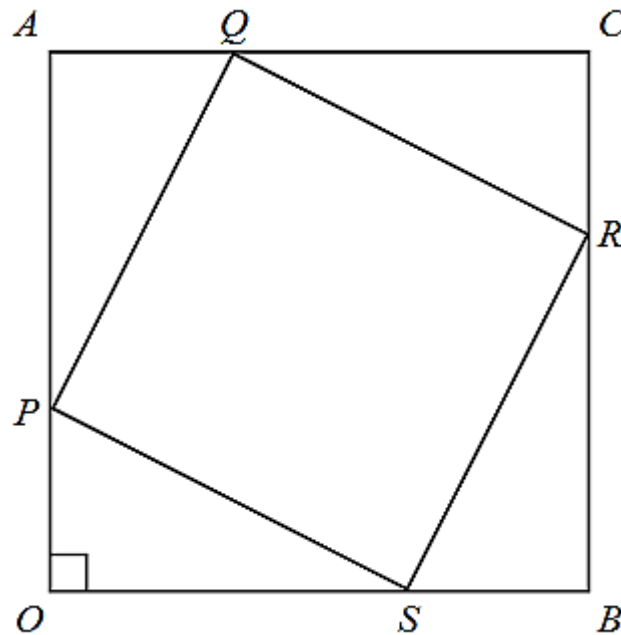
Question 14 continues on page 15

Question 14 (continued)

- (d) Consider square $OACB$ where point O is the origin.

Let the position vector for points A, B be defined as $\underline{a}$ and $\underline{b}$ respectively.

Let the points P, Q, R and S be defined so that $\overrightarrow{OP} = k\underline{a}$, $\overrightarrow{AQ} = k\underline{b}$, $\overrightarrow{RC} = k\underline{a}$ and $\overrightarrow{SB} = k\underline{b}$ where $0 \leq k \leq 1$. This means that points P, Q, R and S are positioned along their respective sides in equal proportion.



- i. Using vector methods, prove that $\angle PQR = 90^\circ$ 3
- ii. Now suppose that in square $OACB$, it is known that $OA = 10$ cm and that point P 3
is moving away from the origin at a speed of 0.2 cm per second.

This means that points Q, R and S are moving at the same speed along their respective sides.

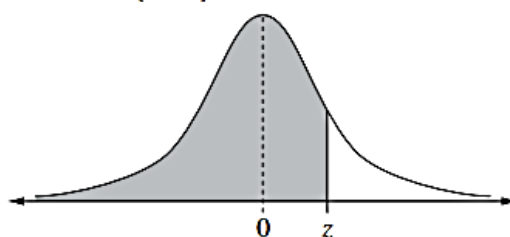
Let the distance of OP be x .

Determine the rate at which the area of square $PQRS$ is changing when $x = 3$ cm.

End of examination

Use the information below to answer **Question 13 (d)** and **Question 14 (b)**

Table of values $P(Z \leq z)$ for the normal distribution $N(0, 1)$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995

2026 Mathematics Extension 1

Marking Guidelines

Section I

Multiple – choice Answer Key

Question	Answer
1	C
2	D
3	A
4	C
5	A
6	D
7	D
8	C
9	A
10	C

Section II

Question 11 (a)

Criteria	Marks
Correctly state the inverse for the restricted domain	2
Correctly find the inverse function, $f^{-1}(x) = 4 \pm \sqrt{x+3}$	1

Sample answer:

$$y = (x - 4)^2 - 3$$

$$x = (y - 4)^2 - 3$$

$$x + 3 = (y - 4)^2$$

$$\pm\sqrt{x+3} = y - 4$$

$$y = 4 \pm \sqrt{x+3}$$

$\therefore f^{-1}(x) = 4 - \sqrt{x+3}$ because we have restricted the domain for $x \leq 4$.

Marker's feedback:

Question was done well by all students

Question 11 (b)

Criteria	Marks
Provide the correct solution	2
Correctly recognise the number of ways to arrange the group of vowels in alphabetical order	1

Sample answer:

There are $\binom{7}{3}$ different arrangements for the vowels in alphabetical order and $\frac{4!}{2!}$ arrangements for the remaining letters. So the total number of arrangements are $\binom{7}{3} \times \frac{4!}{2!} = 420$.

Marker's feedback:

Question was done poorly by most students. The most common mistake was that students assumed the vowels are together, hence leading to the incorrect answer.

Question 11 (c)

Criteria	Marks
Provide the correct solution	2
Correctly applying the chain rule or differentiating $\cos^{-1}(4x^2)$	1

Sample answer:

$$f(x) = (\cos^{-1}(4x^2))^2$$

$$f'(x) = 2 \times \frac{-8x}{\sqrt{1-16x^4}} \times \cos^{-1}(4x^2)$$

$$f'(x) = -\frac{16x \cos^{-1}(4x^2)}{\sqrt{1-16x^4}}$$

Marker's feedback:

Question was done well by all students.

Question 11 (d)

Criteria	Marks
Provide the correct solution	2
Recognise that $\tilde{v}$ is double the magnitude of $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$, or equivalent merit	1

Sample answer:

$\text{proj}_{\mathbf{v}} \mathbf{u} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ and its magnitude is 5. Since $|\tilde{v}| = 10$.

$\tilde{v}$ is parallel to $\mathbf{u}$ and double the magnitude.

$$\therefore \tilde{v} = \begin{pmatrix} 8 \\ -6 \end{pmatrix}$$

Marker's feedback:

Question was poorly done by all students. The question can be done easily by first finding the magnitude of vector $\text{proj}_{\mathbf{v}} \mathbf{u}$ and recognising that $\tilde{v}$ is parallel to $\mathbf{u}$ and double the magnitude.

Students who received one mark either use the dot product to simplify their expression or found the magnitude of vector $\text{proj}_{\mathbf{v}} \mathbf{u}$.

Question 11 (e)

Criteria	Marks
Provide the correct solution	2
Recognise and correctly convert $\sin^2(3u + 1)$ into $\frac{1}{2}(1 - \cos^2 2(3u + 1))$	1

Sample answer:

$$\int \sin^2(3u + 1) du$$

$$= \int \frac{1}{2}(1 - \cos 2(3u + 1)) du$$

$$= \frac{1}{2} \int 1 - \cos(6u + 2) du$$

$$= \frac{1}{2}u - \frac{1}{12}\sin(6u + 2) + C$$

Marker's feedback:

Question was done well by all students

Question 11 (f) (i)

Criteria	Marks
Provide the correct solution	1

Sample answer:

$$\overrightarrow{PR} = \overrightarrow{OR} - \overrightarrow{OP}$$

$$\therefore \overrightarrow{PR} = \underset{\sim}{r} - \underset{\sim}{p}$$

Marker's feedback:

Question was done well by all students

Question 11 (f) (ii)

Criteria	Marks
Provide correct solution	2
Recognise and also correctly apply that $\overrightarrow{RM} = \frac{1}{3}\overrightarrow{RQ}$	1

Sample answer:

$$\begin{aligned}
 \overrightarrow{RM} &= \frac{1}{4}\overrightarrow{RQ} \\
 &= \frac{1}{4}(\overrightarrow{OQ} - \overrightarrow{OR}) \\
 &= \frac{1}{4}(\mathbf{\tilde{q}} - \mathbf{\tilde{r}})
 \end{aligned}$$

Marker's feedback:

Question was done well by all students

Question 11 (f) (iii)

Criteria	Marks
Provide correct solution	2
Recognise and also correctly apply that $\overrightarrow{OM} = \overrightarrow{OR} + \overrightarrow{RM}$	1

Sample answer:

$$\begin{aligned}
 \overrightarrow{PM} &= \overrightarrow{OM} - \overrightarrow{OP} \\
 &= (\overrightarrow{OR} + \overrightarrow{RM}) - \overrightarrow{OP} \\
 &= \underset{\sim}{r} + \frac{1}{4}(\underset{\sim}{q} - \underset{\sim}{r}) - \underset{\sim}{p} \\
 &= \frac{3}{4}\underset{\sim}{r} + \frac{1}{4}\underset{\sim}{q} - \underset{\sim}{p}
 \end{aligned}$$

Marker's feedback:

Question was done well by all students

Question 12 (a) (i)

Criteria	Marks
Using the correct initial values to provide the correct solution	2
Using separation of variables correctly to get $P(t) = 200 - Be^{-kt}$	1

Sample answer:

$$\frac{dP}{dt} = k(200 - P)$$

$$\int \frac{1}{200 - P} dP = \int k dt$$

$$-\ln|200 - P| = kt + C, \text{ where } C \text{ is a constant}$$

$$\ln|200 - P| = -kt + C$$

$$|200 - P| = e^{-kt+C}$$

$$|200 - P| = Ae^{-kt}, \text{ where } A \text{ is a constant}$$

$$200 - P = Be^{-kt}, \text{ where } B \text{ is a constant}$$

$$P = 200 - Be^{-kt}$$

$$\text{Since } t = 0, P(0) = 20$$

$$\therefore B = 180$$

$$\therefore P = 200 - 180e^{-kt}$$

Marker's feedback:

Question was done well by all students

Question 12 (a) (ii)

Criteria	Marks
Provide the correct solution	2
Find the correct exact value of $k = -\frac{1}{10} \ln\left(\frac{49}{60}\right)$	1

Sample answer:

Since $P(10) = 53$,

$$200 - 180e^{-10k} = 53$$

$$\therefore k = -\frac{1}{10} \ln\left(\frac{49}{60}\right)$$

$$\therefore P(20) = 200 - 180e^{-20k}$$

$$= 79.95\text{ }^{\circ}\text{C}$$

$$= 80\text{ }^{\circ}\text{C}$$

Marker's feedback:

Question was done well by all students

Question 12 (b) (i)

Criteria	Marks
Provide the correction solutions with the correct value of α	3
Correctly find that $R = \sqrt{13}$ and an equivalent value of α but not in the correct domain	2
Correctly simplifying $f(x)$ into $3 \cos x - 2\sqrt{3} \sin x$	1

Sample answer:

$$\begin{aligned}
 f(x) &= 3 \sin\left(x + \frac{\pi}{2}\right) - 4 \cos\left(\frac{\pi}{3} - x\right) \\
 &= 3\left(\sin x \cos \frac{\pi}{2} + \sin \frac{\pi}{2} \cos x\right) - 4\left(\cos \frac{\pi}{3} \cos x + \sin \frac{\pi}{3} \sin x\right) \\
 &= \cos x - 2\sqrt{3} \sin x
 \end{aligned}$$

$$\begin{aligned}
 R &= \sqrt{1^2 + (2\sqrt{3})^2} \\
 &= \sqrt{13}
 \end{aligned}$$

$$\sqrt{13} \cos(x + \alpha) = \sqrt{13} \cos x \cos \alpha - \sqrt{13} \sin x \sin \alpha$$

$$\therefore \sqrt{13} \cos \alpha = 1 \quad -\sqrt{13} \sin \alpha = -2\sqrt{3}$$

$$\cos \alpha = \frac{1}{\sqrt{13}} \quad \sin \alpha = \frac{2\sqrt{3}}{\sqrt{13}}$$

$$\therefore \alpha = 1.3 \text{ rad}$$

Marker's feedback:

Question was done well by all students.

Question 12 (b) (ii)

Criteria	Marks
Provide the correct minimum value of $f(x)$ between 0 and 2π	2
Find a minimum value of $f(x)$ but not necessary between 0 and 2π	1

Sample answer:

Since $f(x) = \sqrt{13} \cos(x + 1.3)$,

Then $f'(x) = -\sqrt{13} \sin(x + 1.3)$

To find the minimum value, let $f'(x) = 0$

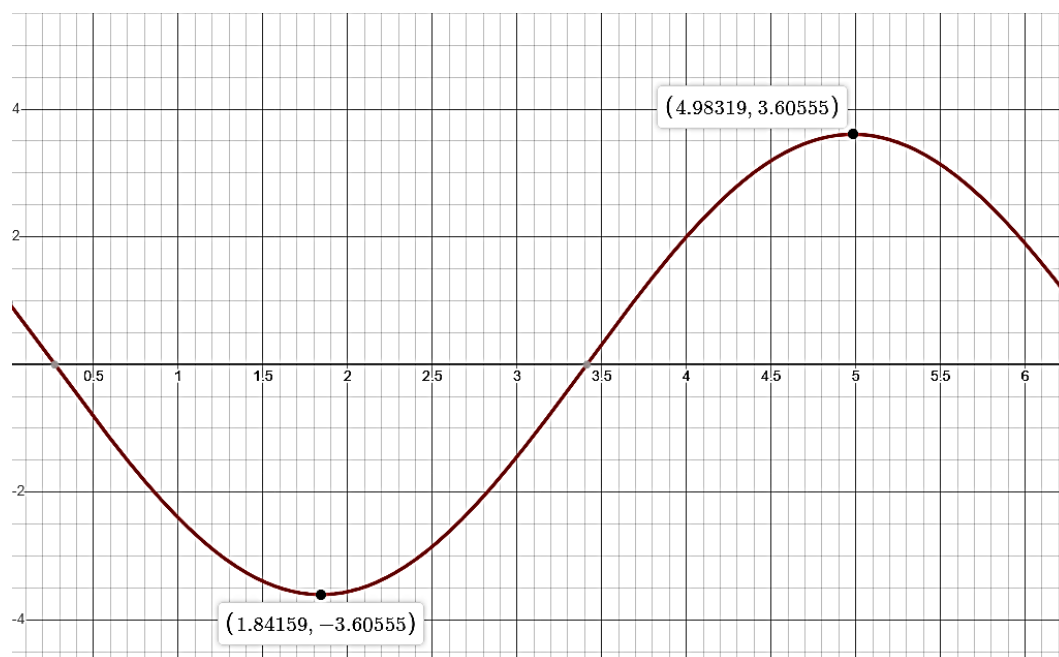
$$-\sqrt{13} \sin(x + 1.3) = 0$$

$$\sin(x + 1.3) = 0$$

$$x + 1.3 = 0, \pi, 2\pi$$

$$x = -1.3, \pi - 1.3, 2\pi - 1.3$$

$$x = -1.3, 1.84, 4.98$$



From the graph, the minimum value is $x = 1.84$ for $0 \leq x \leq 2\pi$

Marker's feedback:

Question was done well by all students

Question 12 (c)

Criteria	Marks
Provide the correct solution	2
Recognise the total number of chords that can be formed with any two points on the circle or recognise the total combination of four chords	1

Sample answer:

Total number of chords that can be formed with any two points on the circle: $\binom{6}{2}$

Total combination of four chords: $\binom{15}{4}$

A quadrilateral can be formed in $\binom{6}{4}$ ways

$$\frac{\binom{6}{4}}{\binom{15}{4}} = \frac{1}{91}$$

Marker's feedback:

Question was poorly done by all students. Many students were only able to find the total number of chords that can be formed with any two points on the circle.

Question 12 (d)

Criteria	Marks
Provide correct solutions	3
Integrate $(x + 2)^4$ or $\frac{1}{(x+2)^2}$ correctly	2
Recognise the correct limit for integration and set up the correct integral to calculate volume	1

Sample answer:

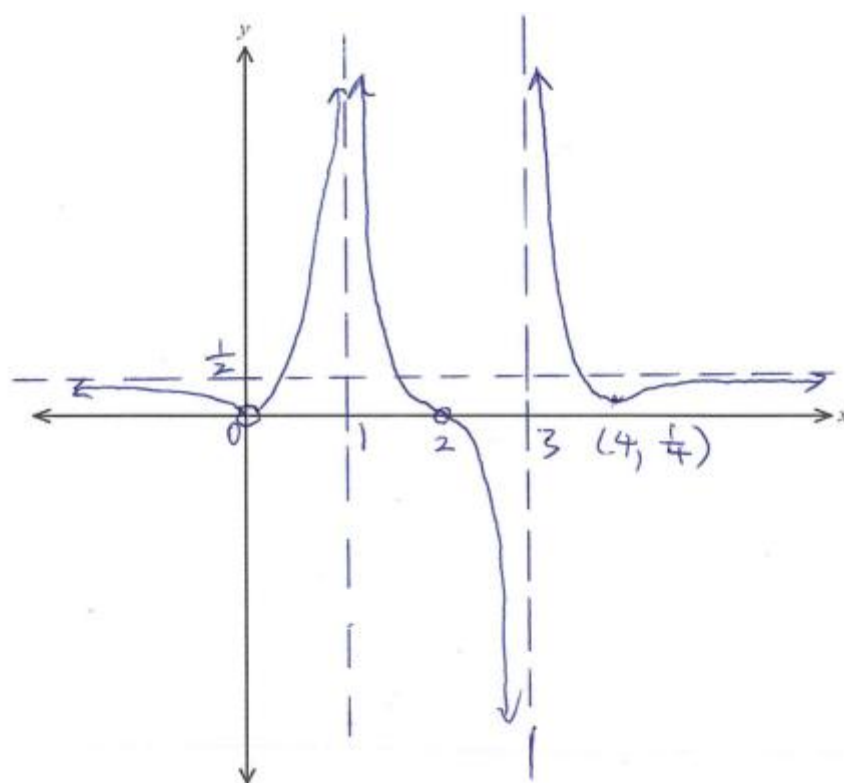
$$\begin{aligned}
 \text{Volume} &= \pi \int_{-1}^1 (x + 2)^4 dx - \pi \int_{-1}^1 \frac{1}{(x + 2)^2} dx \\
 &= \pi \left[\frac{(x + 2)^5}{5} \right]_{-1}^1 - \pi \left[\frac{(x + 2)^{-1}}{-1} \right]_{-1}^1 \\
 &= \frac{242\pi}{5} - \frac{2\pi}{3} \\
 &= \frac{716\pi}{15}
 \end{aligned}$$

Marker's feedback:

Question was done well by all students

Question 13 (a)

Criteria	Marks
Provide the correct solution	3
Clearly show the points of discontinuity at $x = 0$ and $x = 1$	2
Correct asymptotes and correct turning point at $(4, 2)$	1

Sample answer:**Marker's feedback:**

Many students missed the open circle for $y=0$

Question 13 (b)

Criteria	Marks
Provide the correct solution	2
Considered the identity $(1+x)^n(1+x)^n = (1+x)^{2n}$ and showed correct relevant progress to showing the required statement	1

Sample answer:

Consider the following identity: $(1+x)^n(1+x)^n = (1+x)^{2n}$

Using the Binomial Theorem expansion to the LHS:

$$\left[x^n + \binom{n}{1}x^{n-1} + \binom{n}{2}x^{n-2} + \dots + \binom{n}{n} \right] \left[x^n + \binom{n}{1}x^{n-1} + \binom{n}{2}x^{n-2} + \dots + \binom{n}{n} \right]$$

Using the Binomial Theorem expansion to the RHS:

$$x^{2n} + \binom{2n}{1}x^{2n-1} + \binom{2n}{2}x^{2n-2} + \dots + \binom{2n}{2n}$$

Let $x = 1$ to both LHS and RHS:

$$\therefore \left(1 + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \dots + \binom{n}{n} \right)^2 = 1 + \binom{2n}{1} + \binom{2n}{2} + \dots + \binom{2n}{2n}$$

Marker's feedback:

Where students had the right idea, setting out was often poor.

Question 13 (c)

Criteria	Marks
Provide the correct solution	3
Correctly simplifying $\frac{u^3}{1+u}$ using polynomial long division or equivalent merit	2
Correctly finding the new integral and limits in terms of u	1

Sample answer:

$$\begin{aligned}
 & \int_0^{16} \frac{x}{1+\sqrt{x}} dx & \text{If } x=0, u=0 & & u=\sqrt{x}, \\
 & = \int_0^4 \frac{u^2}{1+u} 2\sqrt{x} du & \text{And } x=16, u=4 & & \frac{du}{dx} = \frac{1}{2\sqrt{x}} \\
 & = 2 \int_0^4 \frac{u^3}{1+u} du & & & dx = 2\sqrt{x} du
 \end{aligned}$$

Using polynomial long division

$$\begin{aligned}
 & = 2 \int_0^4 \left(u^2 - u + 1 - \frac{1}{u+1} \right) du \\
 & = 2 \left[\frac{u^3}{3} - \frac{u^2}{2} + u - \ln|u+1| \right]_0^4 \\
 & = \frac{104}{3} - 2 \ln 5
 \end{aligned}$$

Marker's feedback:

Most did not see to use the polynomial division to split the fraction, and so got stuck at this point.

Question 13 (d) (i)

Criteria	Marks
Provide the correct solution	1

Sample answer:

$$\begin{aligned}\sigma &= \sqrt{\frac{pq}{n}} \\ &= \sqrt{\frac{0.25 \times 0.75}{1000}} \\ &= 0.0137\end{aligned}$$

Marker's feedback:

Largely well done.

Question 13 (d) (ii)

Criteria	Marks
Provide the correct solution by using the standard normal distribution table	2
Calculate the correct z score	1

Sample answer:

$$E(\hat{p}) = 0.25 \text{ and } \sigma = 0.0137$$

$$z = \frac{0.23 - 0.25}{0.0137} = -1.4598 \dots \approx -1.46$$

$$P(z \geq -1.46) = P(z \leq 1.46)$$

$$= 0.9276$$

Marker's feedback:

Largely well done, some concern students did not understand the context of their answer.

Question 13 (e)

Criteria	Marks
Provide the correct solution	3
Make correct progress towards showing that $b^3 = ca^3$	2
Establish the correct Vieta formulas for the polynomial	1

Sample answer:

The roots of the polynomial are α , αr and αr^2 .

The sum of roots is: $\alpha + \alpha r + \alpha r^2 = -a$

The sum of two roots is: $\alpha^2 r + \alpha^2 r^3 + \alpha^2 r^2 = b$

And the product of roots is: $\alpha^3 r^3 = -c$

$$\alpha + \alpha r + \alpha r^2 = -a \quad - - - 1$$

$$\alpha^2 r + \alpha^2 r^3 + \alpha^2 r^2 = b \quad - - - 2$$

$$\alpha^3 r^3 = -c \quad - - - 3$$

Divide Equation 1 by Equation 2:

$$\frac{\alpha}{\alpha^2 r} = -\frac{a}{b}$$

Then:

$$\alpha r = -\frac{b}{a}$$

$$\alpha^3 r^3 = -\frac{b^3}{c^3}$$

$$-c = -\frac{b^3}{c^3}$$

$$b^3 = a^3 c$$

Marker's feedback:

Very poorly set out, and many students did not explain their working. Where students established the equations of the roots successfully, many gave no indication of how they used them in their working.

Question 14 (a)

Criteria	Marks
Provide correct solution with correct mathematical reasoning	2
Correctly recognising the number of pigeons and the number of pigeonholes	1

Sample answer:

The pigeonholes represent all the possible positive differences between any pair of integers that are less than or equal to 15. i. e. 1, 2, 3, 4, 5, ..., 14

There are $\binom{8}{2} = 28$ possible combinations of integer pairs, which forms our pigeons.

But for the positive difference of 14, there can only be one pair, i. e. {1, 15}. So now there are only 13 pigeonholes and 27 pigeons. Since $27 = 2 \times 13 + 1$, that means there must be at least three pairs of integers that have the same positive difference.

Marker's feedback:

Not done well. Lot of students could not define what pigeons and pigeonholes represent.

Question 14 (b)

Criteria	Marks
Provide the correct solution	4
Correctly find the corresponding z-score using the z-score table	3
Correctly find values of $E(X)$ and σ	2
Correctly find the value of p	1

Sample answer:

$$n = 100$$

$$\text{Var}(X) = 16$$

$$100p(1 - p) = 16$$

$$25p^2 - 25p + 4 = 0$$

$$\therefore p = 0.8, \text{ since } p > 0.5$$

$$\therefore E(X) = 80 \text{ and } \sigma = 4$$

Since $P(X \geq k) = 0.0668$, then $P(X \leq k) = 0.9332$.

Using the z-score table, k corresponds to a z-score of 1.5.

$$\frac{k - 80}{4} = 1.5$$

$$k = 86$$

Marker's feedback:

Lot of students able to find z-score using the table. However, not many students able to establish quadratic equation in terms of p.

Question 14 (c) (i)

Criteria	Marks
Provide the correct proof	2
Using the correct t-substitution for $\cot\left(\frac{x}{2}\right)$ and $\cot(x)$	1

Sample answer:

$$\text{Let } t = \tan\left(\frac{x}{2}\right)$$

$$\begin{aligned} LHS &= \frac{1}{2} \times \frac{1}{t} - \frac{1}{t} \\ &= \frac{1}{2t} - \frac{1 - t^2}{2t} \\ &= \frac{1 - (1 - t^2)}{2t} \end{aligned}$$

$$\begin{aligned}
&= \frac{t^2}{2t} \\
&= \frac{t}{2} \\
&= \frac{1}{2} \tan\left(\frac{x}{2}\right) \\
&= RHS
\end{aligned}$$

Marker's feedback:

Done well, in general.

Question 14 (c) (ii)

Criteria	Marks
Correct reasoning for $n = k+1$ and correct conclusion.	3
Correct prove for $n = k + 1$	2
Correct prove for $n = 1$ and assumption statement for $n = k$.	1

Sample answer:

When $n = 1$,

$$LHS = \frac{1}{2} \tan\left(\frac{1}{2}\right)$$

$$RHS = \frac{1}{2} \cot\left(\frac{x}{2}\right) - \cot x$$

$\therefore$ True when $n = 1$

Assume true for $n = k$, that is

$$\frac{1}{2} \tan\left(\frac{x}{2}\right) + \frac{1}{4} \tan\left(\frac{x}{4}\right) + \frac{1}{8} \tan\left(\frac{x}{8}\right) + \cdots + \frac{1}{2^k} \tan\left(\frac{x}{2^k}\right) = \frac{1}{2^k} \cot\left(\frac{x}{2^k}\right) - \cot x$$

Now prove true for $n = k + 1$, that is to prove:

$$\frac{1}{2}\tan\left(\frac{x}{2}\right) + \frac{1}{4}\tan\left(\frac{x}{4}\right) + \cdots + \frac{1}{2^k}\tan\left(\frac{x}{2^k}\right) + \frac{1}{2^{k+1}}\tan\left(\frac{x}{2^{k+1}}\right) = \frac{1}{2^{k+1}}\cot\left(\frac{x}{2^{k+1}}\right) - \cot x$$

$$LHS = \frac{1}{2^k}\cot\left(\frac{x}{2^k}\right) - \cot x + \frac{1}{2^{k+1}}\tan\left(\frac{x}{2^{k+1}}\right), \text{ by assumption}$$

$$= \frac{1}{2^k}\cot\left(\frac{x}{2^k}\right) - \cot x + \frac{1}{2^k}\left(\frac{1}{2}\tan\left(\frac{x}{2^{k+1}}\right)\right)$$

$$= \frac{1}{2^k}\cot\left(\frac{x}{2^k}\right) - \cot x + \frac{1}{2^k}\left(\frac{1}{2}\cot\left(\frac{x}{2^{k+1}}\right) - \cot\left(\frac{x}{2^k}\right)\right), \text{ using part (i)}$$

$$= \frac{1}{2^k}\cot\left(\frac{x}{2^k}\right) - \cot x + \frac{1}{2^{k+1}}\cot\left(\frac{x}{2^{k+1}}\right) - \frac{1}{2^k}\cot\left(\frac{x}{2^k}\right)$$

$$= \frac{1}{2^{k+1}}\cot\left(\frac{x}{2^{k+1}}\right) - \cot x = RHS$$

Hence the statement is also true for $n = k + 1$. So the statement is true for all positive integers by mathematical induction.

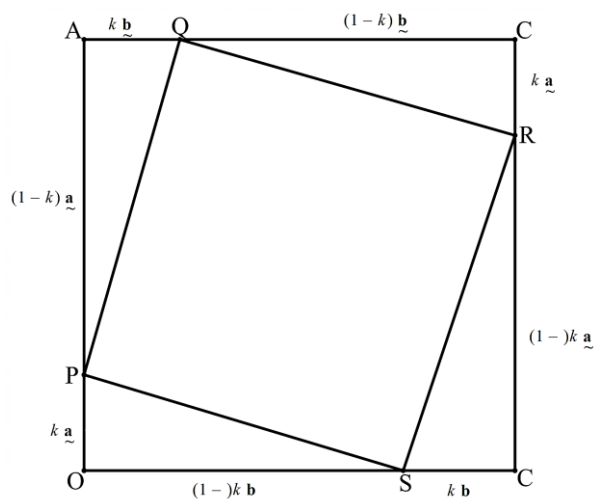
Marker's feedback

Most of students could prove $n = 1$. Some students were able to recognise using part (i) to prove for $n = k + 1$.

Question 14 (d) (i)

Criteria	Marks
Provide the correct solution	3
Recognise and apply the dot product on the correct vectors	2
Find the correct expression for $\overrightarrow{PQ}$ and $\overrightarrow{QR}$ or equivalent merit	1

Note: One mark deducted if there is no vector notation presented throughout the working.

Sample answer:

$$\overrightarrow{PQ} = (1-k)\underline{a} + k\underline{b} = \underline{a} - k\underline{a} + k\underline{b}$$

$$\overrightarrow{QR} = (1-k)\underline{b} - k\underline{a} = \underline{b} - k\underline{b} + k\underline{a}$$

Using dot product:

$$\overrightarrow{PQ} \cdot \overrightarrow{QR} = \left((1-k)\underline{a} + k\underline{b} \right) \cdot \left((1-k)\underline{b} - k\underline{a} \right)$$

$$= (1-k)^2 \underline{a} \cdot \underline{b} - k(1-k) \left| \underline{a} \right|^2 + k(1-k) \left| \underline{b} \right|^2 + k^2 \underline{a} \cdot \underline{b}$$

Since $\underline{a} \cdot \underline{b} = 0$ and $\left| \underline{a} \right| = \left| \underline{b} \right|$, then

$$\overrightarrow{PQ} \cdot \overrightarrow{QR} = 0$$

Marker's feedback:

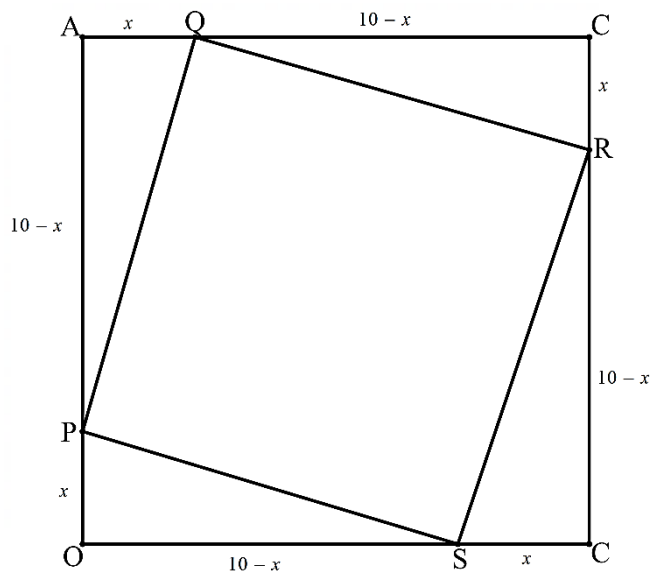
Not done well. The working layout done poorly, some students treated vector PQ and vector QR as position vectors.

Question 14 (d) (ii)

Criteria	Marks
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Provide the correct solution	3
Set up the correct related rates of change equation	2
Find the length of PQ and the area of the square $PQRS$	1

Sample answer:



$PQ = \sqrt{2x^2 - 20x + 100}$ using pythagoras' theorem.

Therefore, the area, $A = 2x^2 - 20x + 100$.

Then $\frac{dA}{dx} = 4x - 20$

$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$$

$$= (4x - 20) \times 0.2$$

$$= -1.6 \text{ cm}^2 \text{ s}^{-1}$$

$\therefore$ The area of the square is decreasing by $1.6 \text{ cm}^2 \text{ s}^{-1}$

Marker's feedback:

A few students got full marks. Lots of students recognised using differentiation related rates, however, quite a few students could not find correct length of PQ .